Tropical Cyclones activity outlook

SARCOF-33

Laurent Labbé & Sébastien Langlade

Météo-France / Regional center for indian Ocean

laurent.labbe@meteo.fr

Southern Africa Regional Climate Outlook Forum
August 27 2026
- Swakopmund - Namibia -

SARCOF-33 : TC activity forecast - 2026/08

1 – Climatology

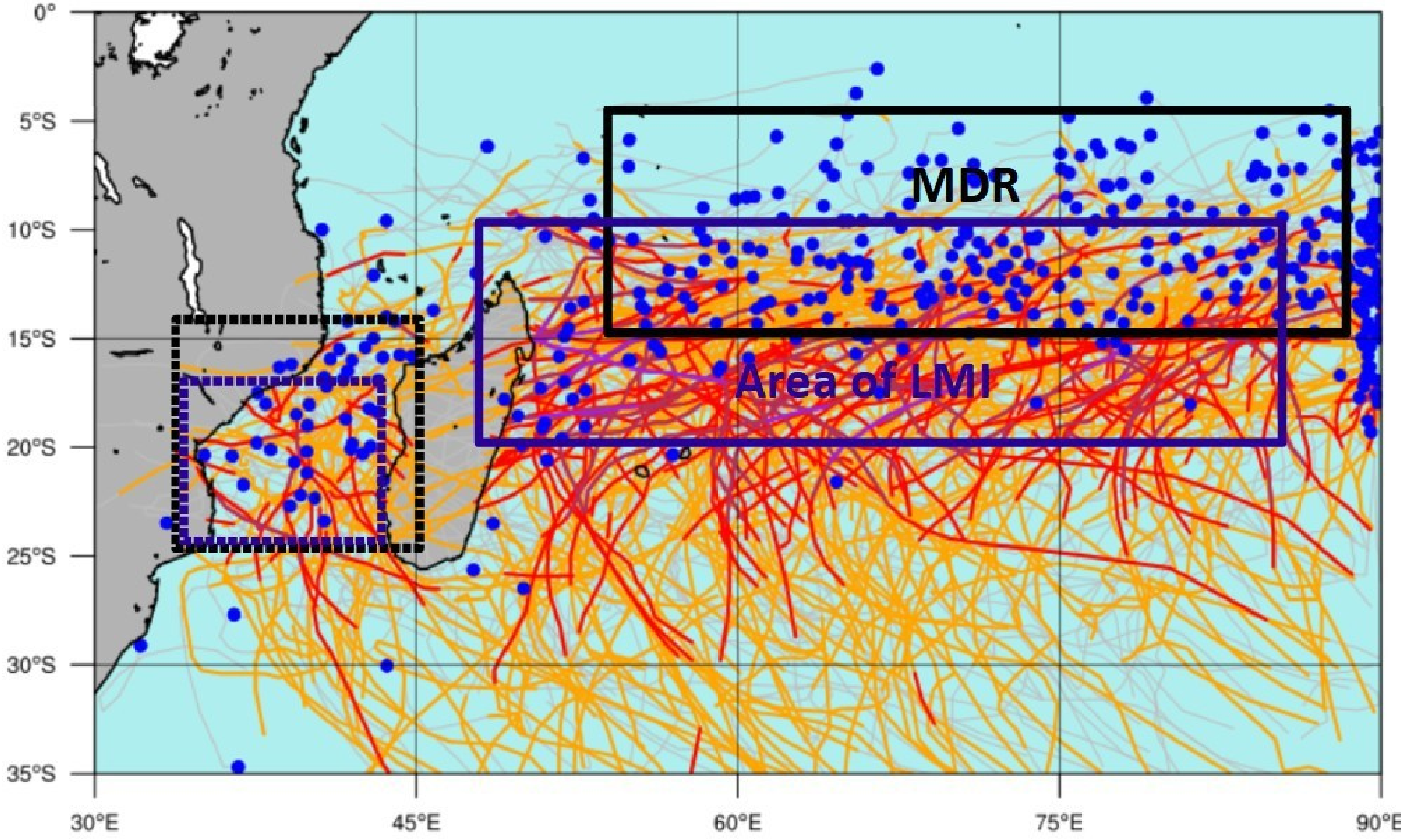
2 – Climate drivers impact on TC activity

3 – Preliminary outlook

This presentation shows the first guess concerning the tropical cyclones activity for the upcoming rainy season. Due to the uncertainties remaining at this stage, the outlook will contain mainly qualitative information. The final forecast will be issued by the RSMC in late october.

1 – TC climatology

Climatology of Tropical Storms-Cyclones (1986-2022)



- **Main Development Region**: 10°S ± 5° and east of 55°E

- **Life Maximum Intensity** usually reached between 10°S and 20°S east of Madagascar

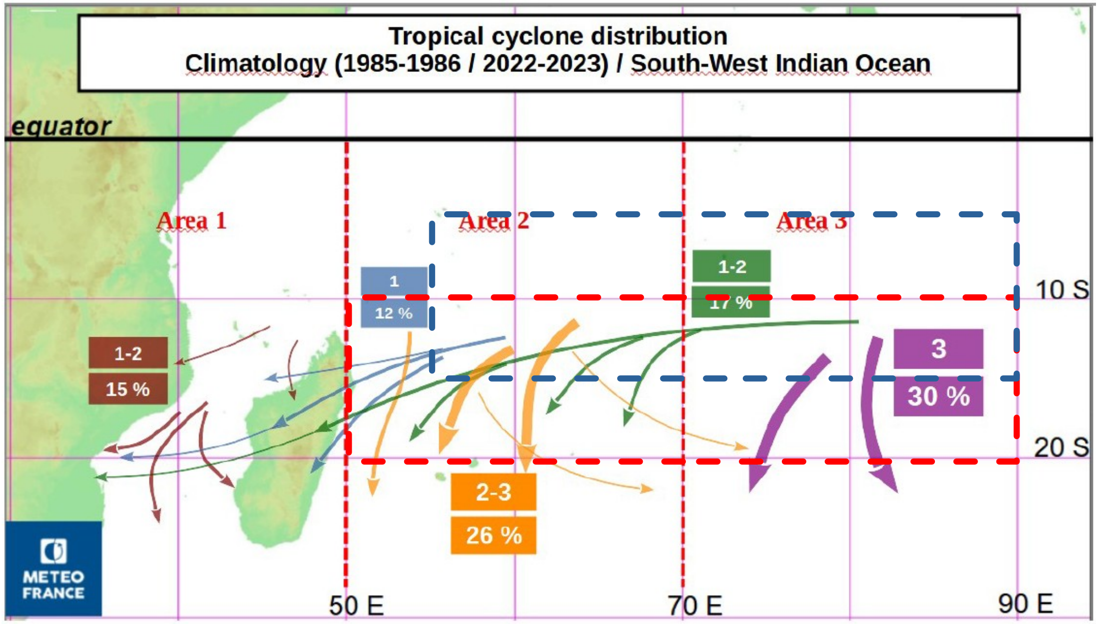
- **Mozambique Channel** : ~15 % of TC geneses with a local MDR and LMI area shifted southward

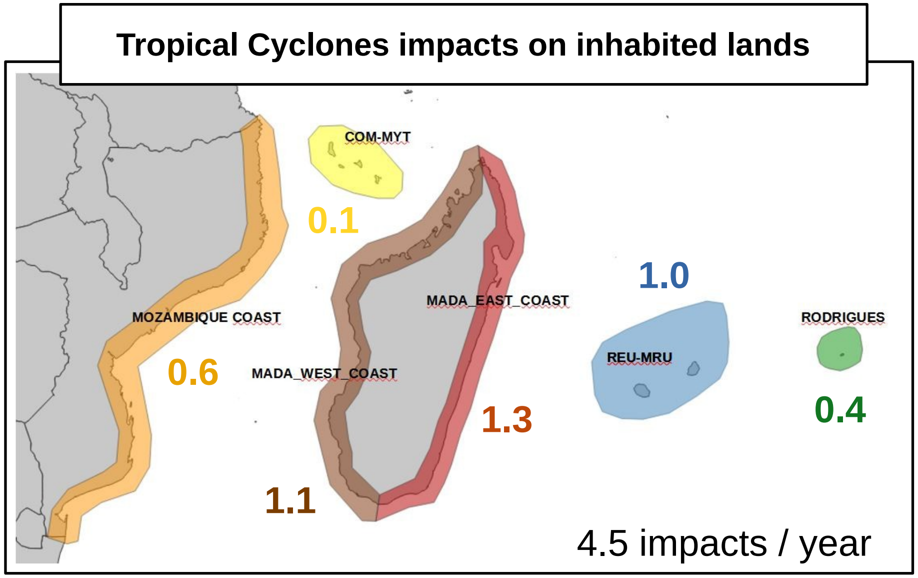
- In average, 4 to 5 TCs per season, bring at least moderate rain and/or wind impacts over inhabited lands

1 – TC climatology

The distribution of TC tracks can be splitted into different catégories depending on :

- Their development area (3 zones)
- The main orientation of their track (zonal-parabolic / meridian)



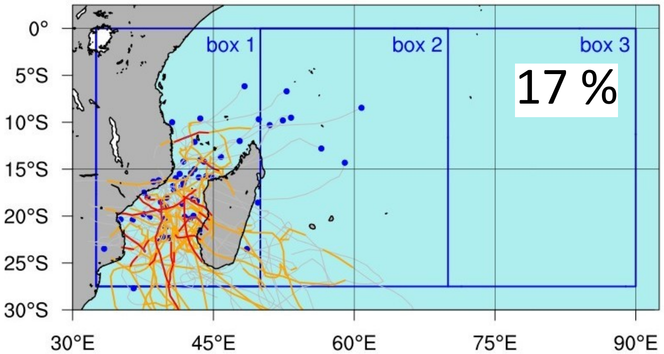


This observed distribution is associated to a risk of impact over inhabited lands

	Tropical storms-cyclones > 33kt	Tropical cyclones > 63kt	Intense tropical cyclones > 89kt	Very intense tropical cyclones > 115kt
Average annual numbers	9,8	5,3	3,3	0,6
Ratio	100 %	54 %	34 %	6 %

1 – TC climatology

Track classe : 111 (1986-2022)

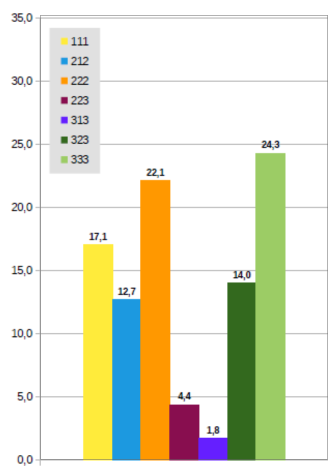


Classification of tracks with respect to :

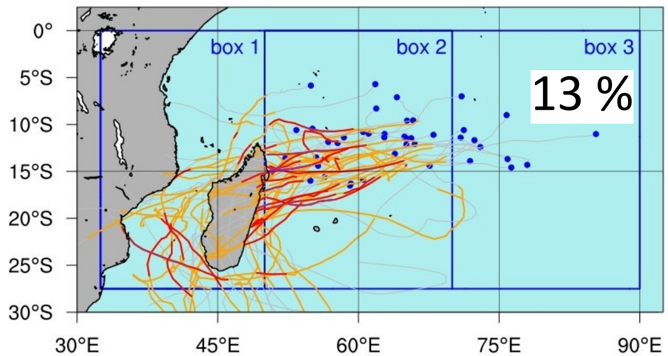
- start longitude (box 1,2,3)
- min longitude (box 1,2,3)
- max longitude (box 1,2,3)

with :
Vmax ≥ 34kt (10 minutes avg wind)
25°S ≤ latitude ≤ 0°

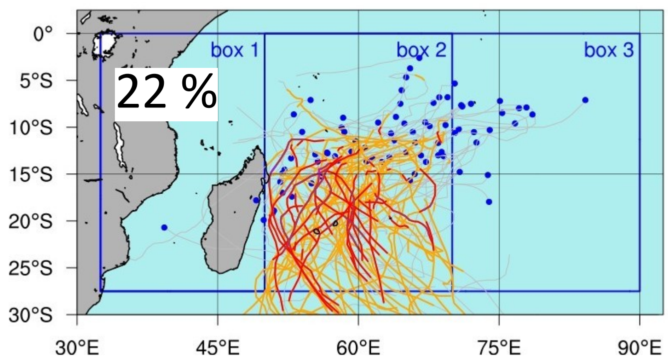
Frequency associated to each track class



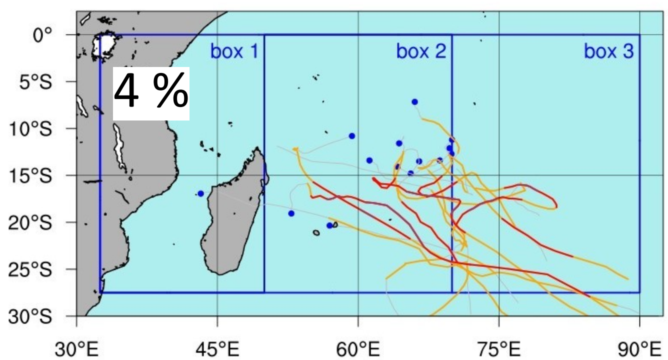
Track classe : 212 (1986-2022)



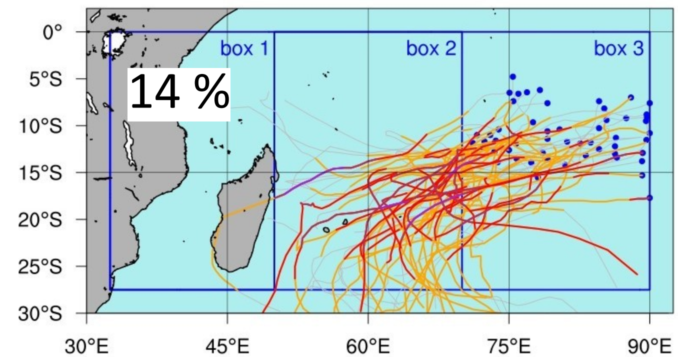
Track classe : 222 (1986-2022)



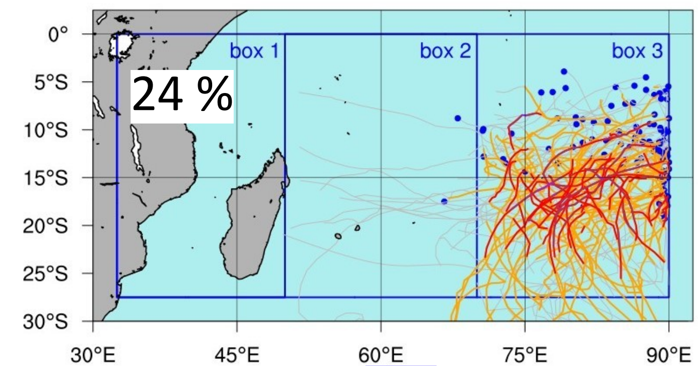
Track classe : 223 (1986-2022)



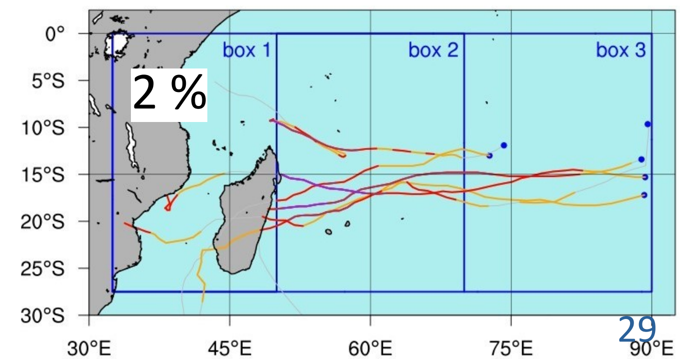
Track classe : 323 (1986-2022)



Track classe : 333 (1986-2022)

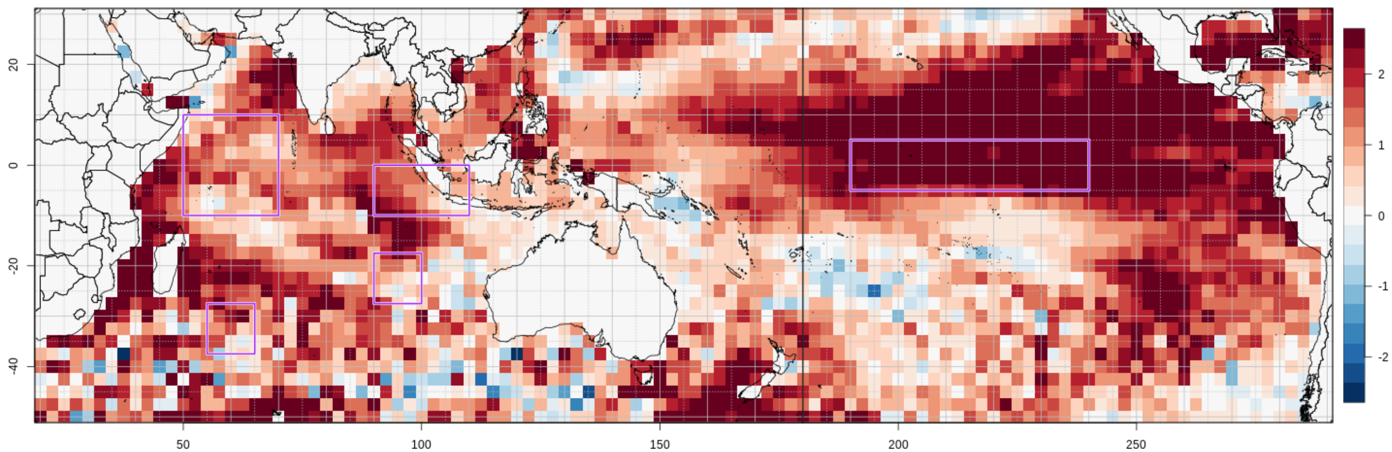


Track classe : 313 (1986-2022)

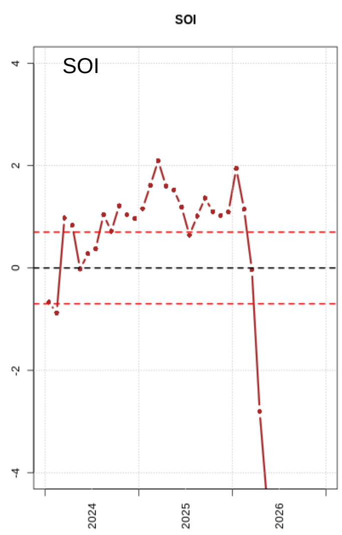


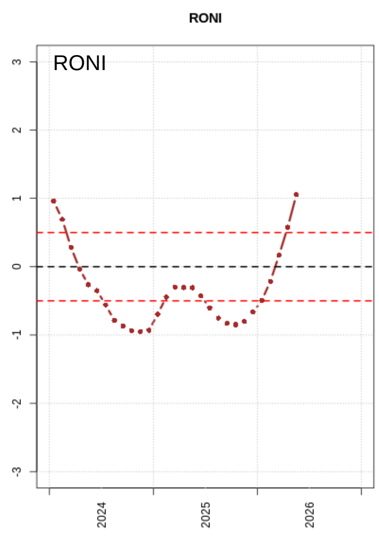
The climate drivers present status

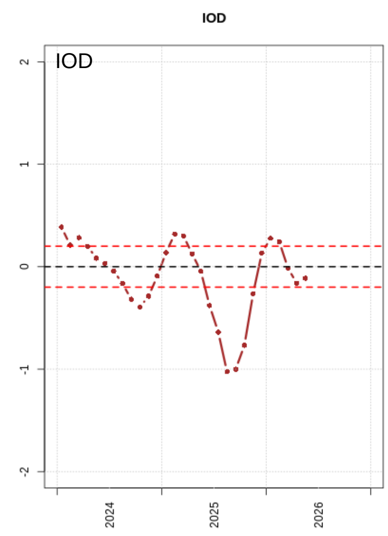
***Situation** of SST std anomaly for the quarter MJJ 2026 (ERA5)*

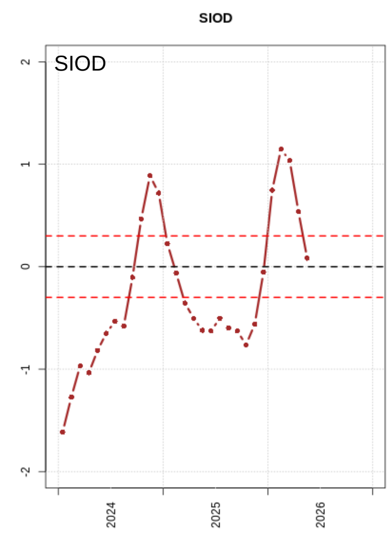


Recent evolution of climate drivers indexes (until MJJ 2026)





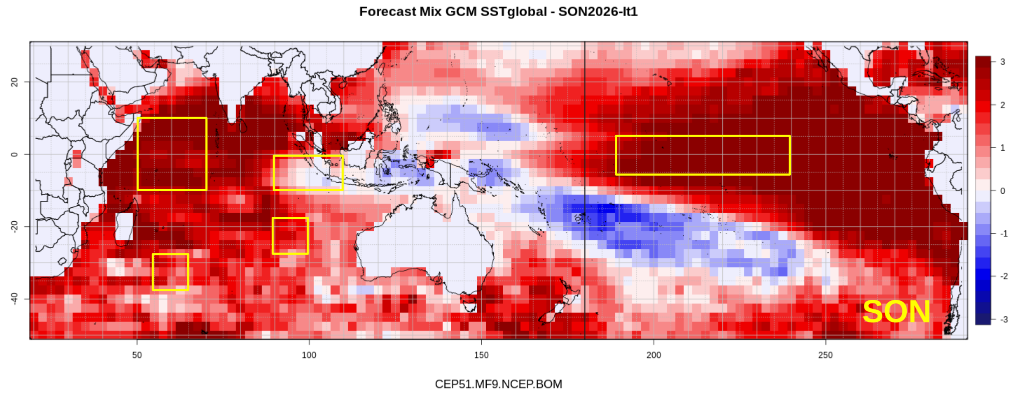


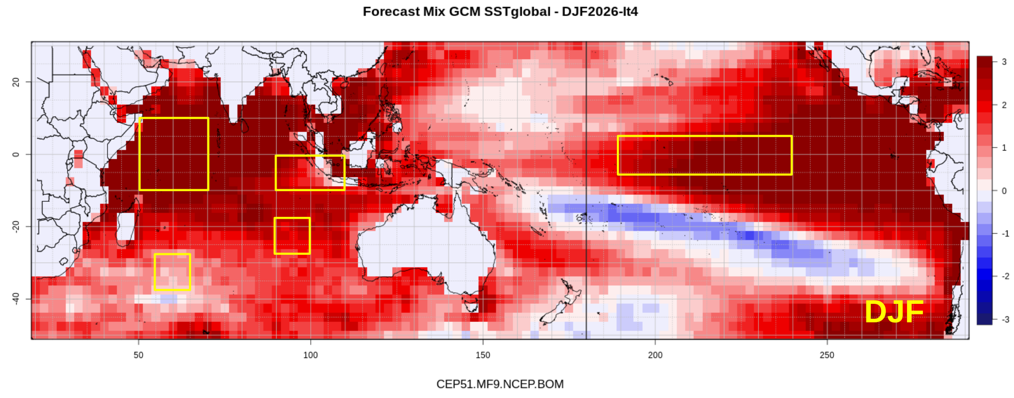


2 – Climate drivers impact

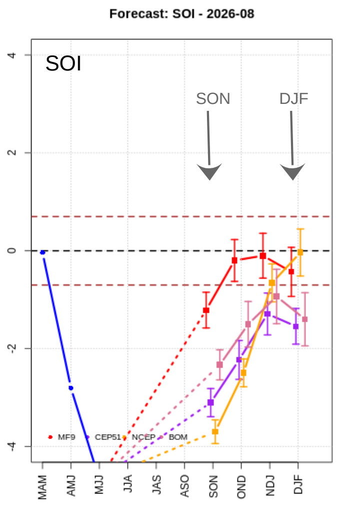
The climate drivers forecast

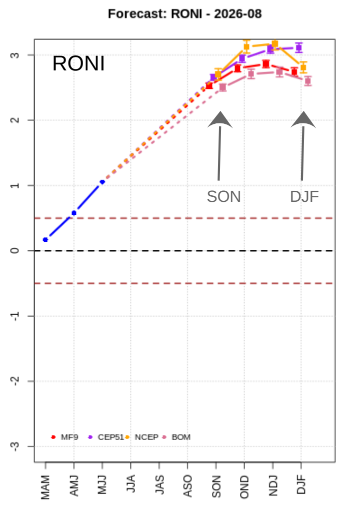
Forecast of SST std. anomaly for SON and DJF 2026 (ECMWF – MF – NCEP - BOM)

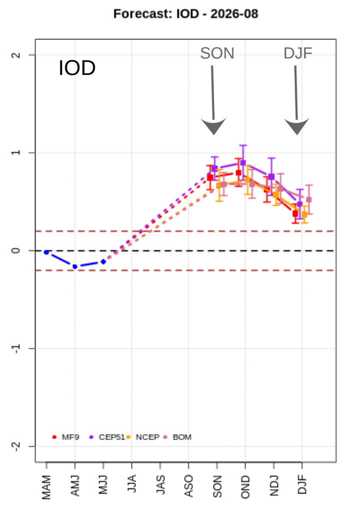


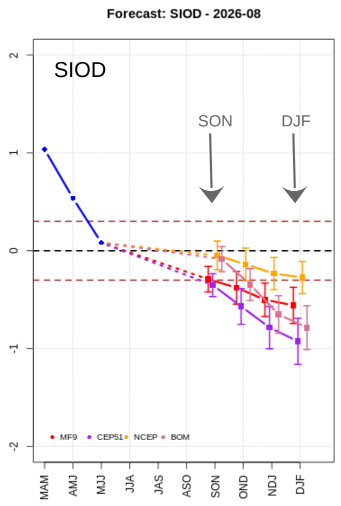


Climate indices multi-model forecast based on august 2026







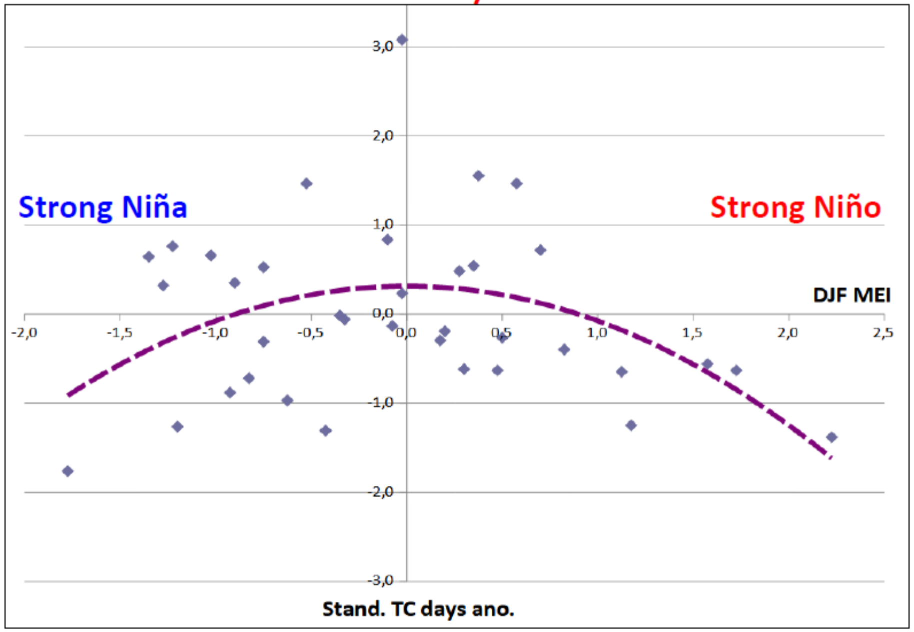


2 – Climate drivers impact

Impact of ENSO on TC activity

Strong negative or positive phases of ENSO have a negative influence on the cyclonic activity. However this signal is far from being systematic since other factors may have an influence this activity, namely drivers like IOD and SIOD.

ENSO is also expected to influence the orientation of the TC trajectories. In a El Nino situation the number of systems showing a meridian orientation increases whereas those showing a zonal orientation decrease. The result of this effect directly impacts the territories which are prone to direct cyclonic hit.



ZONE	**SWIO**	ROD	REU-MAU	MADA EAST COAST	MADA WEST COAST	MOZ	COM
Climatology	**4,5**	0,4	1,0	1,3	1,1	0,6	0,1
NINO	**3,1**	0,3	0,9	**0,9**	**0,7**	**0,4**	0
NINA	**5,9**	0,3	0,8	**2,0**	**1,4**	**1,3**	0,1

2 – Climate drivers impact

Impact of El Nino on TC tracks

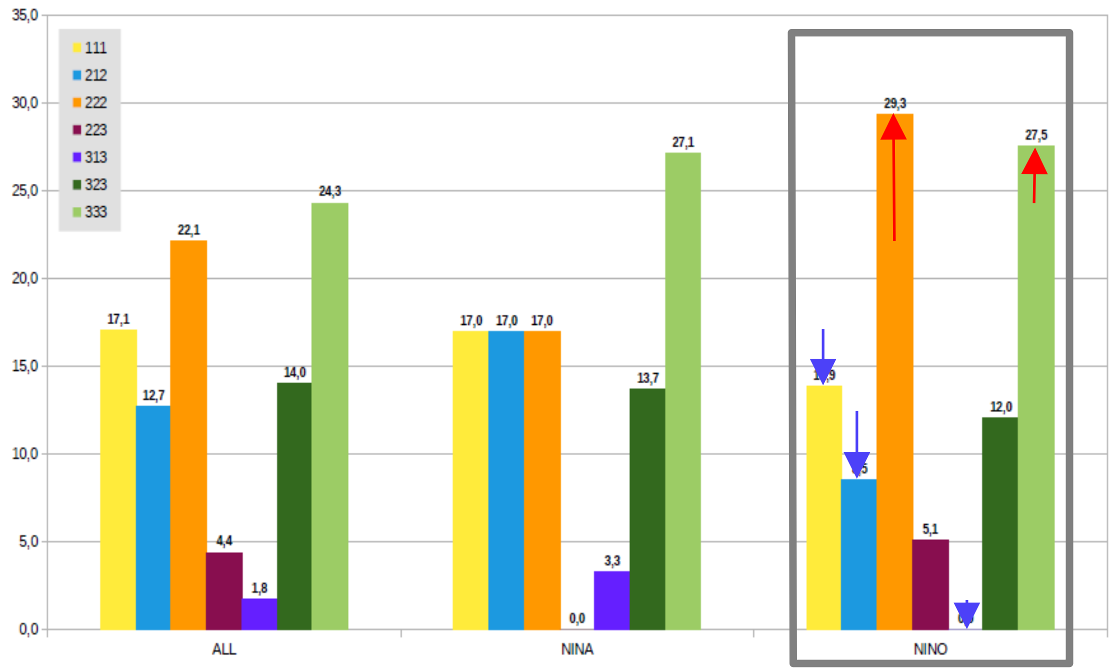
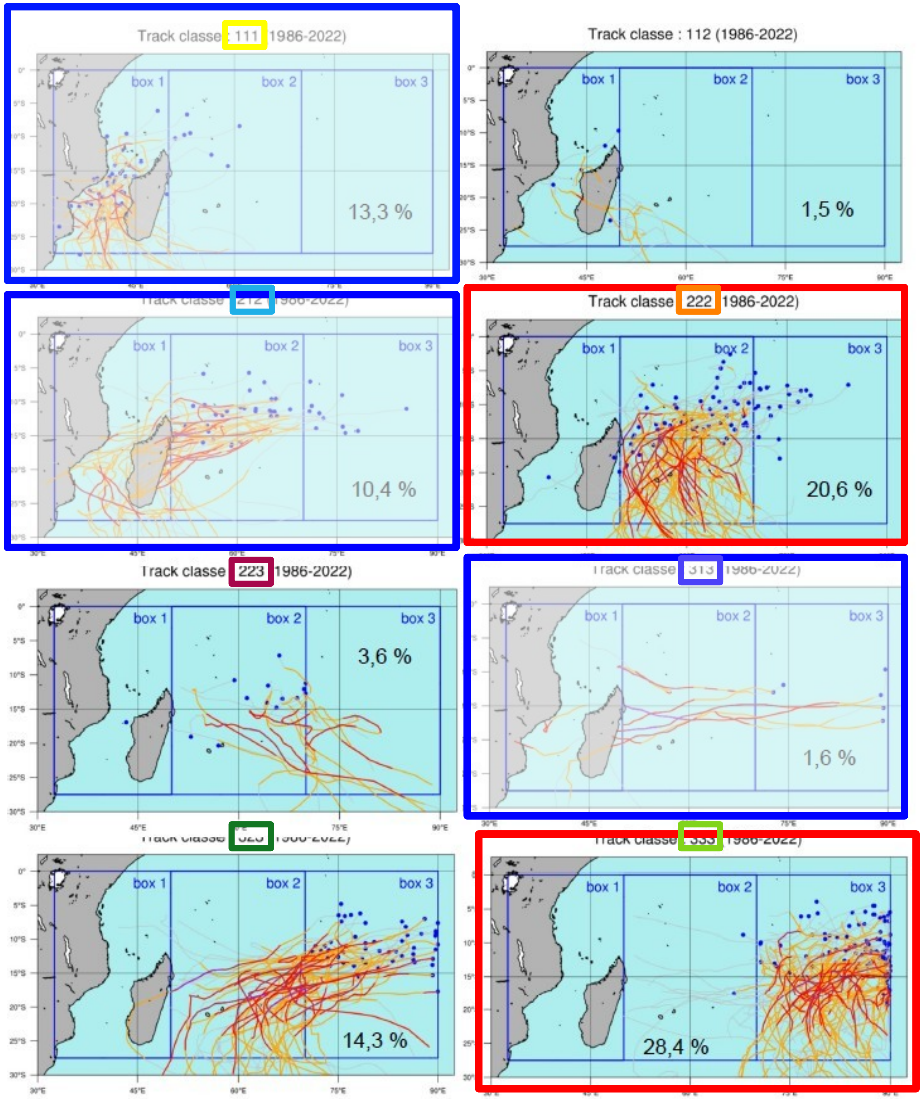


Figure 11 : Modulation of classes proportion with respect to ENSO phases during the core of the cyclonic season (December to March). Proportions are expressed in %



ENSO+ (El Nino)

222, 333 favored

212, 313, 111 reduced

2 – Climate drivers impact

Impact of SIOD- on TC tracks

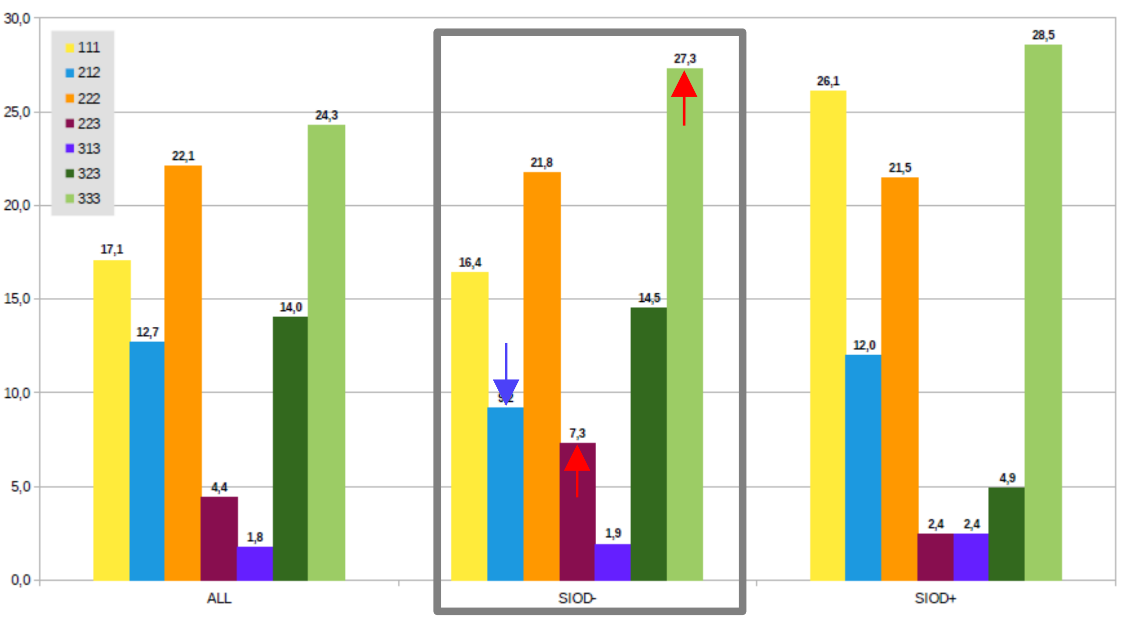
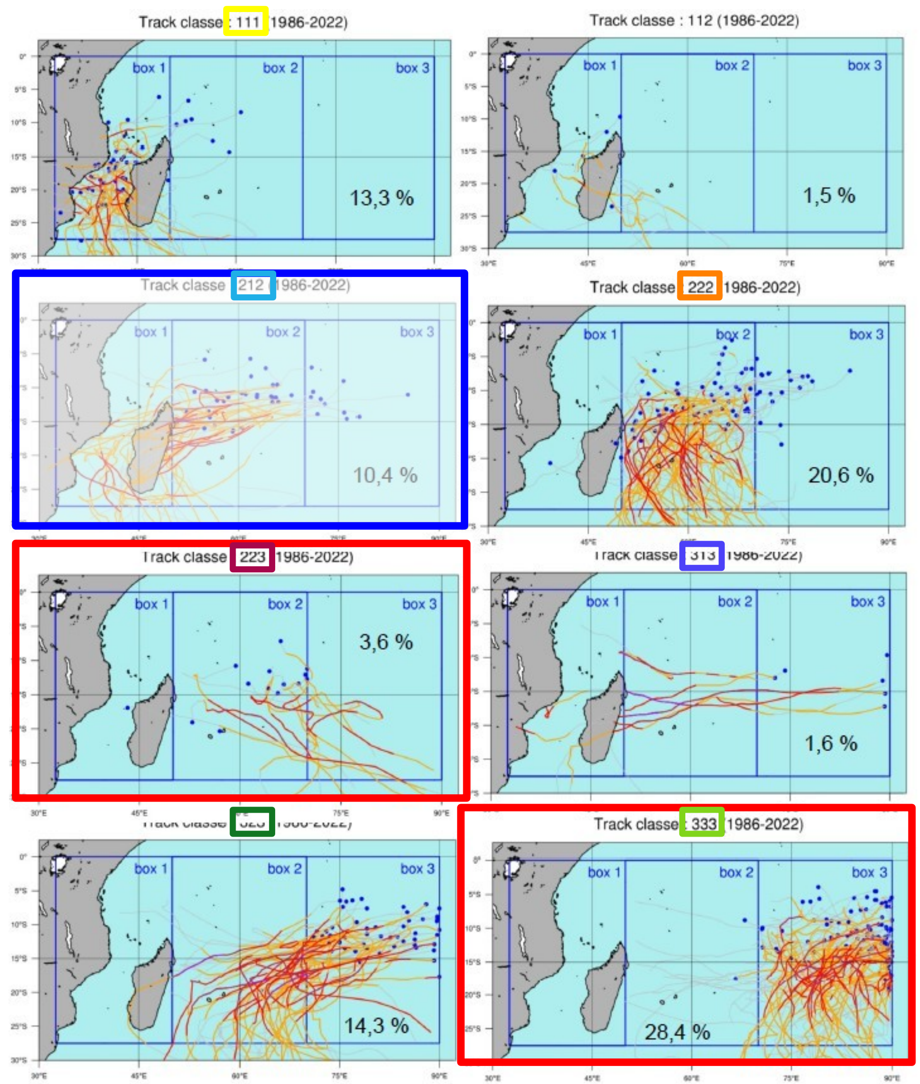


Figure 12 : Modulation of classes proportions with respect to SIOD phases during the core of the cyclonic season (December to March). Proportions are expressed in %

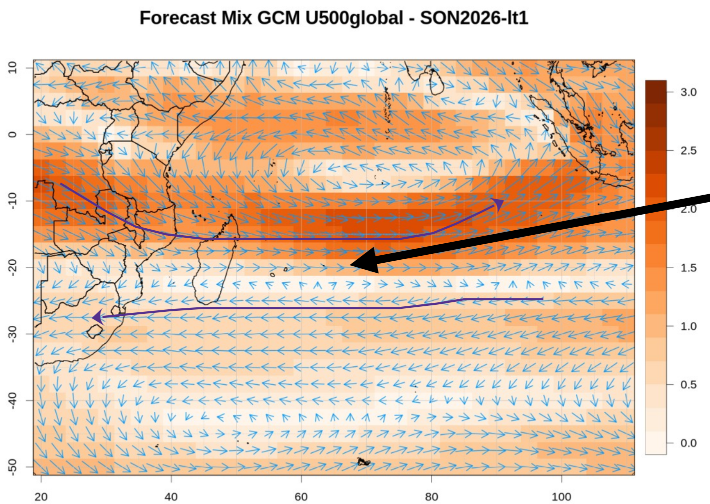


SIOD -
223, **333** favored
212 reduced

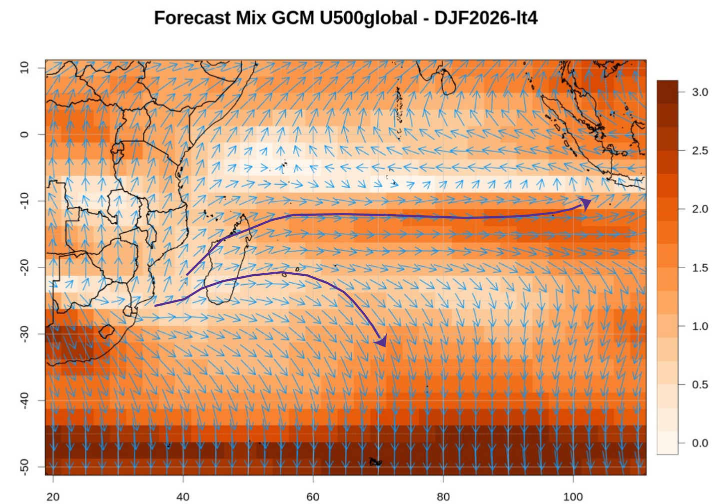
3 – Preliminary outlook

GCM forecast : low level (850hPa) and steering level (500hPa) fluxes

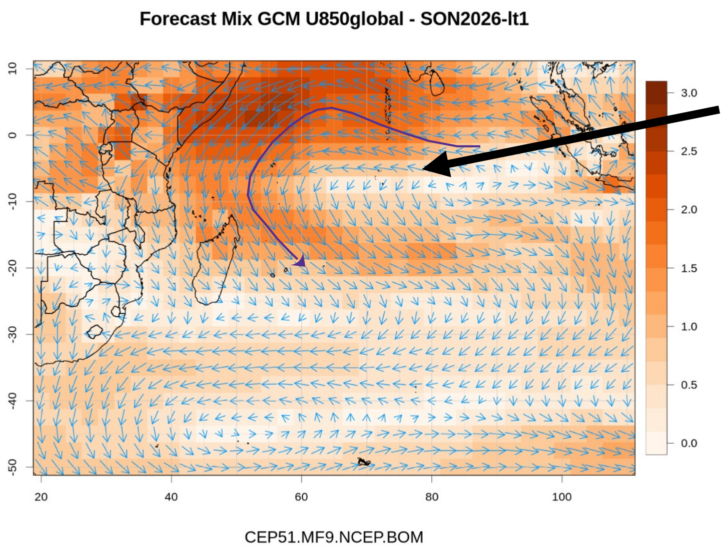
Forecast of 500hPa wind anomaly for SON and DJF 2026 (ECMWF – MF – NCEP - BOM)



Cyclonic shear

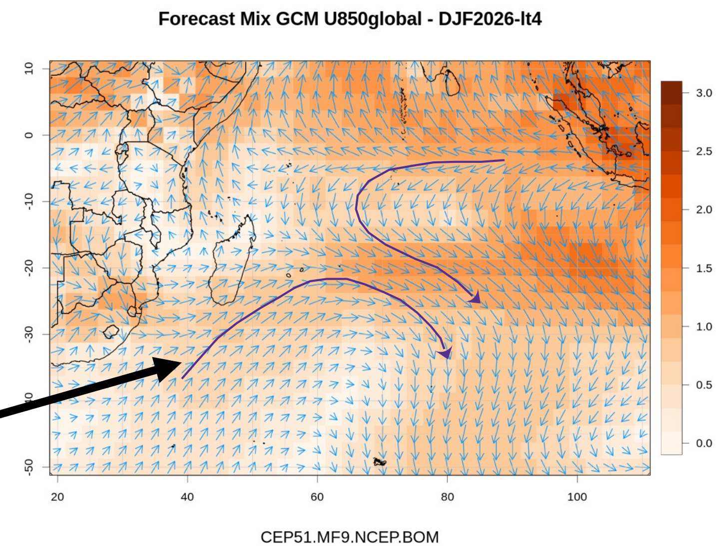


Forecast of 850hPa wind anomaly for SON and DJF 2026 (ECMWF – MF – NCEP - BOM)



ElNino / IOD+

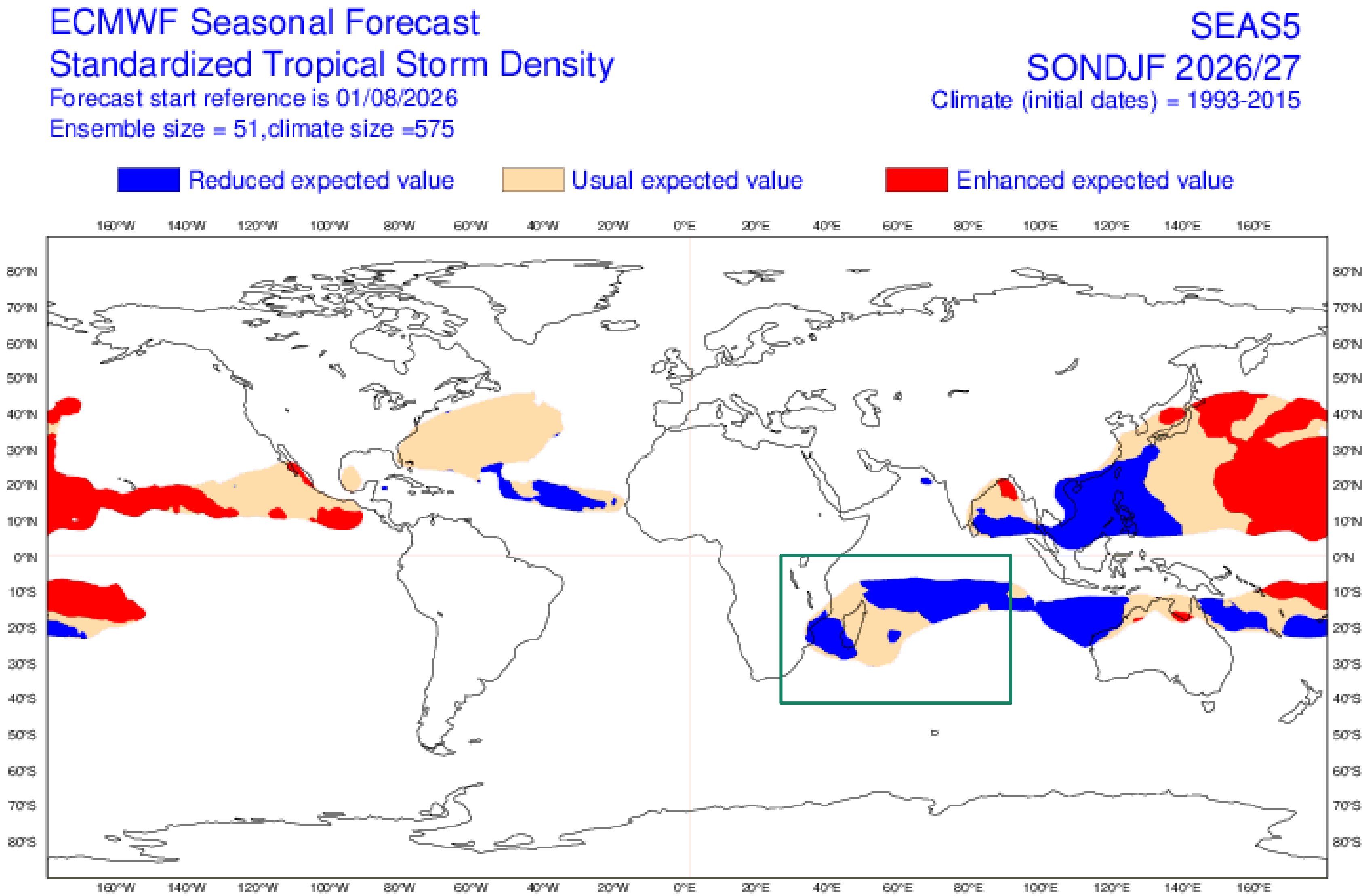
SIOD-



SARC

3 – Preliminary outlook

ECMWF forecast : Tropical storm density



3 – Preliminary outlook

The synthesis of the climatological knowledge and the predicted large scale conditions over the coming rainy season allow to draw some preliminary hints on the expected TC activity :

- Forthcoming Cyclonic season (Nov 2026 to April 2027) :
  - Mainly driven by strong El Nino + positive IOD + negative SIOD configuration
  - Below to strongly below average activity
  - Southerly to South-easterly tracks favored
  - Risk over SWIO countries below normal, particularly for Madagascar and the continent
- Activity during the second half of the season (Feb to Apr). Too early to say! Will depend on the evolution of the climate drivers and particularly SIOD.

This outlook should be updated in the coming months. The TC outlook mini-forum, organized online by PIROI and WMO on the 28th october, will provide a final outlook.


Thank you for listening